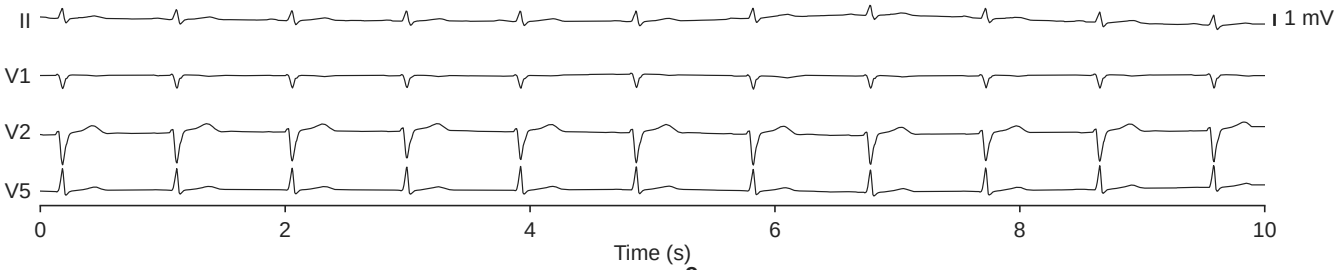
a

Input: d0043.csv, 5,000 rows × 13 columns (t_ms and 12 leads in mV, raw, 500 Hz; 4 of 12 leads shown)
Message (excerpt): "… Patient: age 25 years, male. …"



b

Code run 2 of 4: Python script written by the model
(shell wrapper omitted; long lines cut)

```
import pandas as pd,numpy as np
from scipy.signal import butter,filtfilt,find_peaks
d=pd.read_csv('d0043.csv');fs=500
L=list(d.columns[1:]);X=d[L].values
b,a=butter(2,[0.5,40],btype='band',fs=fs);Y=filtfilt(b,a,X,a …
e=np.sum(np.abs(np.diff(Y,axis=0)),axis=1)
pk,_=find_peaks(e,height=np.percentile(e,98)*0.4,distance=in …
# refine on lead II abs
rr=np.diff(pk)/fs
print('HR',60/rr.mean(),'RR sd',rr.std(),len(pk))
print(np.round(rr,3))
beats=np.array([Y[p-300:p+400] for p in pk if p>300 and p+40 …
M=np.median(beats,axis=0)
t=np.arange(-300,400)*2
for i,l in enumerate(L):
  m=M[:,i]
  print(l,' '.join(f'{v:.2f}' for v in m[250:450:5]))
np.save('/tmp/M.npy',M)
```

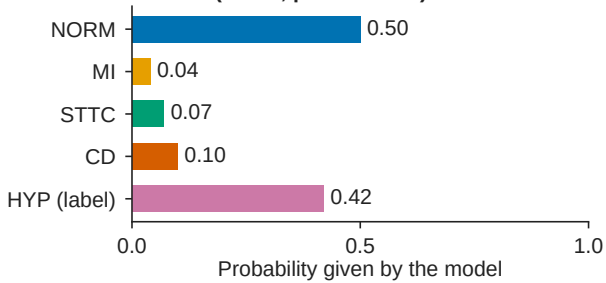
c

Printed output of code run 4

```
on 275 off 324 QRS ms 98
I    R 0.83 S -0.25 J -0.02 J60 0.04 T 0.34
II   R 1.03 S -0.43 J -0.08 J60 0.06 T 0.30
III  R 0.22 S -0.20 J -0.04 J60 0.02 T -0.05
aVR  R 0.34 S -0.93 J 0.06 J60 -0.05 T -0.31
aVL  R 0.31 S -0.04 J 0.01 J60 0.02 T 0.20
aVF  R 0.63 S -0.31 J -0.06 J60 0.03 T 0.13
V1   R 0.14 S -1.30 J 0.02 J60 0.05 T -0.08
V2   R 0.39 S -3.08 J 0.00 J60 0.27 T 0.81
V3   R 0.38 S -1.65 J 0.02 J60 0.28 T 0.79
V4   R 1.41 S -0.73 J -0.05 J60 0.14 T 0.52
V5   R 2.35 S -0.45 J -0.08 J60 0.05 T 0.38
V6   R 1.85 S -0.24 J -0.06 J60 0.02 T 0.30
QT 382 QTc 394.0029160708591
axis 30.67115572484157
```

d

Answer (JSON; probabilities)



e Answer (JSON; key findings)

The rhythm is regular sinus at about 64 bpm, with PR about 150 ms, QRS about 100 ms, QTc about 395 ms, a normal axis near +45° and no pathological Q waves or ST-T abnormality. High precordial voltage (Sokolow-Lyon 3.65 mV = S V1 1.30 + R V5 2.35; S V2 3.1 mV) without strain is voltage-only LVH, which can be a normal finding in a 25-year-old man.

PTB-XL ecg_id 30 (validation fold)
Statements: LVH (likelihood 100), SR.
4 code runs, 2,526 output tokens, USD 0.056 (batch price).